

Episode 56: Once Upon a Time... (37) – The Story of 16mm and 35mm *Tenjou-hen Uchuu Kouji* Part 5

The other day, there was one voicemail on my home phone.

"This is ○○ from Nakano Tax Office. I'm calling about the content of your tax return."

Ugh, the tax office! Actually, this year I did three years' worth of tax returns all at once, and I'd been a bit worried because I hadn't received "OK notifications" from the tax office for the past two years.

Hmm, what could have been wrong? I didn't hide income or fraudulently inflate expenses... I don't think I did... I think I didn't, probably... hmm (sweat). Still a summons? (sweat) ...So I guess I'll get grilled? (heavy sweat)

Nervously, I called the tax office the next day. And then...

"Well, we found calculation errors in three places for fiscal year 2005..."

"Yes"

"Please file an amended return"

"That's all?"

"That's all"

That's all it was (laugh). Since they said "mail is OK," I'll clear it up by having them send documents and responding. Ah, what a relief (laugh). I'm really not good with tax offices (bitter laugh).

Well then.

Around 1990 when we were making *Tenjou-hen Uchuu Kouji*, it was naturally still "cel animation filmed on film," and at that time anime photography used both 16mm and 35mm film.

16mm and 35mm refer to the width of the film. Since film has continuous holes (perforations) on both sides for shooting and projection, the actual image area is even smaller, but back then TV works used 16mm film while theatrical works used 35mm. With larger image area, image density is higher, and 35mm film, being about 4 times the area of 16mm, was naturally suited for large screens like movie theatres. Colour reproduction was also better with 35mm.

Tenjou-hen Uchuu Kouji was shot on this 35mm film. Not just *Tenjou-hen Uchuu Kouji* but all Toei Doga OVA works at the time should have been 35mm. Basically, TV works that aired once and were done, versus video works intended for repeated viewing, were treated like theatrical works.

However, even for TV works, opening and ending films were shot in 35mm, and frequently used bank cuts were often shot in 35mm too. Repeatedly used footage required creating dupe (duplicate) negatives from original film for processing, and 35mm was suited for such things. Nowadays you can just copy and insert on computer during editing (laugh).

So basically, TV work rushes checks were done with 16mm positive film, while theatrical and OVA works were checked with 35mm positive, but...

This 35mm film naturally cost considerably more for both negative film and processing compared to 16mm. So regular rushes checks used 16mm positive film. While shooting itself used 35mm negative film, during printing it was compressed and printed onto 16mm positive film. So what we watched for photography rushes checks was mostly 16mm film footage.

This was quite subtle.

Naturally 16mm used 16mm projectors, and with smaller film screens, the actual light projected onto screens was less, making screens feel darker compared to 35mm rushes. While 16mm was fine for checking animation and camera work itself, colour reproduction and light effects like transmitted light in our domain inevitably looked inferior to 35mm rushes.

However, 16mm had various advantages, primarily being much easier to handle. Not just projection, but especially editing was easier. Editing was naturally done with positive film, and the difficulty between 16mm and 35mm editing was like heaven and earth (though "negative editing" done after all cuts were approved was naturally done with 35mm negative film as the master).

So after checking with 16mm and giving OK, the completed film had final checking with 35mm film.

This final check was called "Print 0 screening," meaning screening with the first positive film processed from the negative assembled through negative editing. Here we checked whether there were any fatal problems. Well, problems would be bad (bitter laugh). I also attended these checks.

What I checked was screen colour reproduction. Film impressions change considerably with printing. Naturally it starts with the negative, but colour tones and transparency can look quite different depending on printing balance.

Also, there were individual differences in cameras and lenses used for shooting—even with the same 35mm cameras, lens individual differences could subtly affect footage colour tones. If everything were shot with identical lenses there'd be no variation, but when shooting in parallel on multiple photography tables, there could often be slight colour variations cut by cut.

Including adjustments for such things, technicians at processing labs would pre-adjust and print Print 0 to some degree (for this, we'd provide combinations of actual cels and backgrounds used in the main story as colour references for each scene), but with animation, since light and colour reproduction aren't shooting natural objects, some effects and colour tones might be better emphasised depending on scenes. When intended effects didn't come out well, we'd have them adjust during printing.

"Make flame colours stronger in this scene" or "make blue sky reproduction clearer," sometimes specifying cuts for adjustments.

After screenings, we'd hold meetings with lab technicians and animation staff including the director to confirm and make requests. These adjustments were called "timing," and we always did this for film-screened works. Even with digital production today, this remains the final checkpoint for theatrical works screened on film.

These Print 0 screenings were held at Toei-affiliated processing lab Toei Chemical Industries (now Toei Lab Tech) in Chofu. For *Tenjou-hen Uchuu Kouji*, roughly every two months per episode. Since other OVA works were also going to Print 0 in parallel, I was going to Toei Chemical quite frequently. On Print 0 days, we'd take 2-3 taxis from Oizumi Studio, and after several trips, I became more familiar with the route from Oizumi to Toei Chemical than the taxi drivers (laugh).

From 1990 through 1991, OVA-centered days continued like this.

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